Performance Test

Date	29 October 2026
Team ID	Individual Project — Ujjawal Rathod
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	

Project Performance Testing:

The project team shall fill the following information in the Project performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Tables Loaded, Records Count and Size Tables: EV_MARKET_TRAIN – 200,000 rows; EV_MARKET_TEST – 100,000 rows; view EV_MARKET_ALL (UNION ALL) – 300,000 rows × 37 columns. `SELECT COUNT(*) FROM EV_MARKET_ALL;  -- 300000` Size: [paste table size from Snowsight > Data > Tables > Table Details] [Insert screenshot: table list with row counts]

2.	Data Preprocessing & Quality	Null checks, duplicate checks, trimming, standardization and data cleaning
		Checks run and findings:
		• Nulls: annual_sales_units ≈ 1%, customer_rating ≈ 2%; nulls also in battery_cycle_life and resale_value_pct_3yr. Kept as NULL – SUM/AVG ignore them.
		• Duplicates: 1 duplicate vehicle_id in 300,000 rows (negligible).
		• Trimming / standardization: TRIM and INITCAP applied to text columns (market_region, category, battery_type, manufacturer); manufacturing_country standardized.
		`SELECT COUNT(*)-COUNT(annual_sales_units) AS null_sales, COUNT(*)-COUNT(customer_rating) AS null_rating FROM EV_MARKET_ALL;`
		`SELECT vehicle_id, COUNT(*) FROM EV_MARKET_ALL GROUP BY vehicle_id HAVING COUNT(*)>1;`
		[Insert screenshot: query results of the checks]

3.	New Fields Created	Any New Fields created • SALES_MILLIONS = annual_sales_units / 1,000,000 • MARKET_SHARE_PCT = SUM(annual_sales_units) / SUM(SUM(annual_sales_units)) OVER () – share of total market (equivalent of the Tableau FIXED LOD). `CREATE VIEW EV_MARKET_ENRICHED AS SELECT *, annual_sales_units/1000000 AS sales_millions FROM`

<table>
<tr><td></td><td></td><td>EV_MARKET_ALL;
[Insert screenshot: new fields in query result]</td></tr>
<tr><td>4.</td><td>Filter Efficiency Test</td><td>Filters used + comparison of query performance with different filters
Filters used: market_region, model_year, battery_type, category (same four as the Tableau dashboard).
<table><tr><th>Filter applied</th><th>Rows returned</th><th>Execution time</th></tr><tr><td>No filter (baseline)</td><td>[ ]</td><td>[ms]</td></tr><tr><td>market_region = one region</td><td>[ ]</td><td>[ms]</td></tr><tr><td>model_year = one year</td><td>[ ]</td><td>[ms]</td></tr><tr><td>battery_type + category</td><td>[ ]</td><td>[ms]</td></tr></table>
[Fill in from Query History after running each query; insert screenshot]</td></tr>
<tr><td>5.</td><td>Dashboard & Visualization Testing</td><td>No of Visualizations / Graphs -
9 visualizations (V1–V9), same as the Tableau dashboard, each backed by a Snowflake SQL query: V1 sales by category; V2 Top 10 manufacturers & market share; V3 sales by model year; V4 battery share; V5 average range by category; V6 range by battery type over years; V7 sales by manufacturing country; V8 rating by category; V9 region × category.
[Insert screenshot: Snowsight dashboard or query results for V1–V9]</td></tr>
<tr><td>6</td><td>Query History</td><td>Query with Highest Execution time
Query with highest execution time: [Query ID / execution time / warehouse – from Snowsight > Monitoring > Query History]
[Insert screenshot: Query History sorted by duration]</td></tr>
</table>